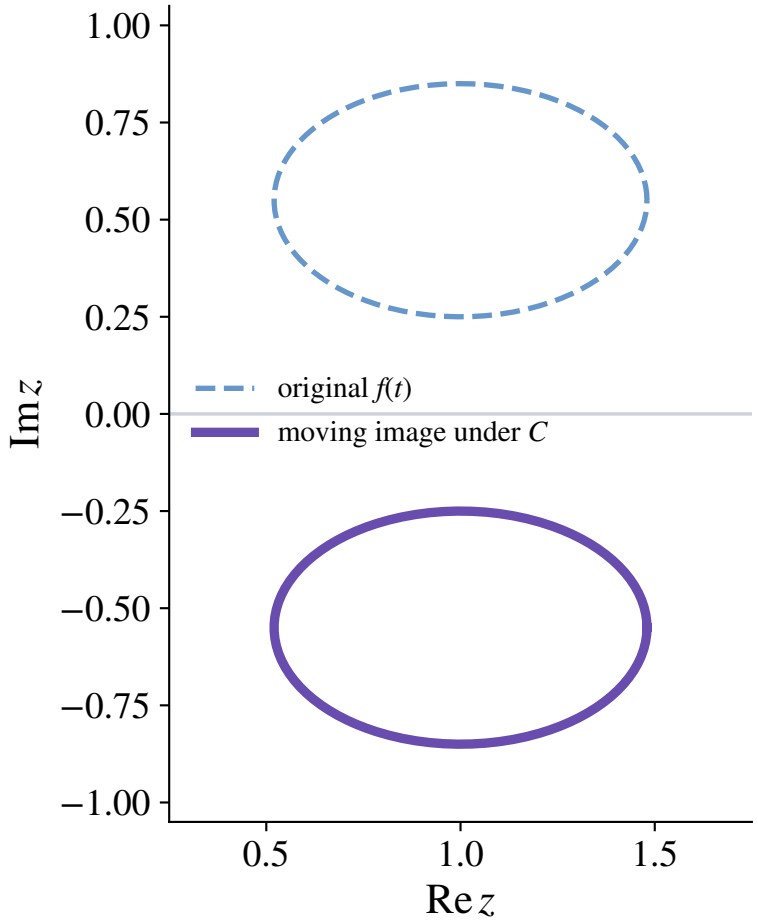
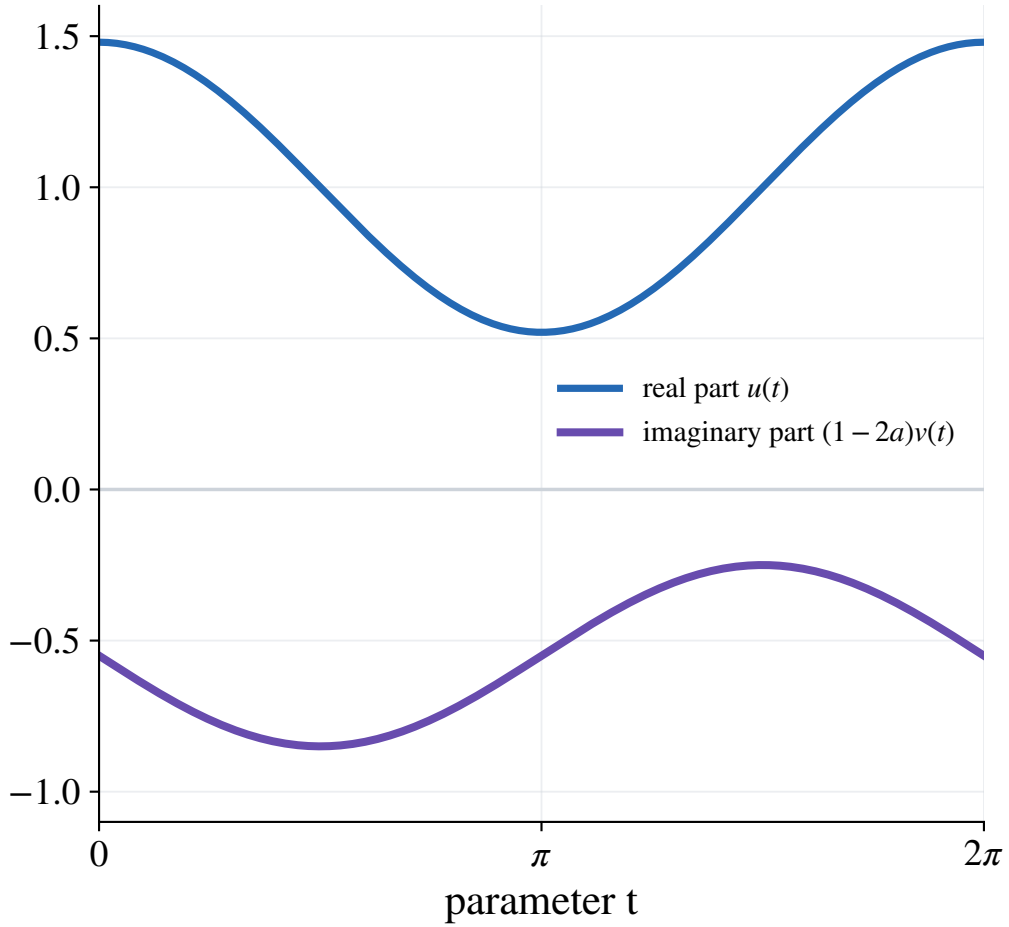
WHY IS COMPLEX CONJUGATION ONLY REAL-LINEAR?

$Cf = \bar{f}$

curve in the complex plane



$f(t) = u(t) + iv(t)$



$C(\lambda f) = \bar{\lambda}\, Cf$

R-linear ✓

C-linear ✗